

Workforce Attrition Patterns and Risk Hotspot Analysis at Palo Alto Networks

Research Paper

Prepared By

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Abstract

Employee attrition is one of the biggest challenges faced by organizations, especially in technology-driven companies like Palo Alto Networks. High employee turnover leads to increased recruitment costs, loss of experienced employees, reduced productivity, and delays in project completion. Understanding the reasons behind employee attrition is essential for creating effective retention strategies.

This project aims to analyze workforce attrition using HR analytics techniques and interactive Power BI dashboards. The analysis identifies the departments, job roles, age groups, and workload factors that contribute most to employee attrition. Python was used for data cleaning, preprocessing, and exploratory data analysis, while Power BI was used to build an interactive dashboard for visualization.

The dashboard enables HR managers to monitor attrition trends using various KPIs and filters. The findings from this analysis help organizations make data-driven decisions to improve employee retention and workforce planning.

Keywords: HR Analytics, Employee Attrition, Power BI, Python, Data Analysis, Dashboard, Palo Alto Networks

1.Introduction

Human Resource Analytics has become an important part of modern organizations. Companies collect large amounts of employee data, but without proper analysis, this data cannot provide meaningful insights. Employee attrition is one of the most important HR metrics because replacing experienced employees requires significant time and cost.

Palo Alto Networks is a leading cybersecurity company where employee knowledge and technical skills are valuable assets. Losing experienced employees can directly impact productivity, innovation, and customer satisfaction.

This project focuses on analyzing employee attrition patterns using historical HR data. Instead of predicting future attrition, the objective is to understand existing attrition trends and identify high-risk employee groups. The project uses Python for data preparation and Power BI for dashboard development.

The dashboard provides HR professionals with a visual representation of employee attrition across departments, job roles, age groups, education fields, overtime status, business travel

frequency, and tenure categories.

2.Problem Statement

Palo Alto Networks has detailed employee information, but HR management lacks clear visibility into workforce attrition trends. Without proper analysis, HR decisions remain reactive rather than proactive.

The organization needs answers to questions such as:

Which departments experience the highest attrition?

Which job roles have the highest employee turnover?

Which age groups are leaving the organization more frequently?

Does overtime increase attrition?

Does business travel influence employee turnover?

Are employees leaving before completing five years in the organization?

Does lack of promotion affect employee retention?

Without these insights, designing effective employee retention strategies becomes difficult.

3 Objectives

The primary objectives of this project are:

Analyze overall employee attrition.

Identify departments with the highest attrition.

Identify job roles with maximum employee exits.

Analyze attrition by age group.

Study gender-wise attrition.

Compare attrition based on marital status.

Analyze education field-wise attrition.

Study overtime impact on attrition.

Analyze business travel impact.

Study promotion stagnation.

Analyze employee tenure patterns.

Build an interactive Power BI dashboard.

Provide recommendations for reducing employee attrition.

4. Dataset Description

The dataset contains HR-related employee information collected from Palo Alto Networks.

Important attributes include:

Age

Attrition

Department

Job Role

Business Travel

Gender

Marital Status

Education Field

Monthly Income

Total Working Years

Years at Company

Years Since Last Promotion

Overtime

Distance From Home

Work-Life Balance

Job Satisfaction

Performance Rating

The dataset was cleaned before analysis to remove inconsistencies and improve data quality.

5.Tools and Technologies

The following tools were used during the project:

Programming Language

Python

Python Libraries

Pandas

NumPy

Matplotlib

Seaborn

Visualization Tool

Power BI

Development Environment

Google Colab

Version Control

GitHub

Documentation

Canva

Microsoft Word

6. Data Cleaning and Preprocessing

Data preprocessing was performed before creating visualizations.

The following steps were completed:

Removed duplicate records.

Checked for missing values.

Corrected inconsistent column names.

Converted data into appropriate formats.

Created Age Groups.

Created Distance Groups.

Created Tenure Groups.

Verified Attrition values.

Checked data quality.

Prepared data for Power BI visualization.

These preprocessing steps improved the reliability of the dashboard and analysis.

7.Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand employee behavior, identify hidden patterns, and discover the major reasons behind workforce attrition.

The following analyses were carried out:

Overall employee attrition rate

Department-wise attrition

Job role-wise attrition

Age group-wise attrition

Gender-wise attrition

Marital status-wise attrition

Education field-wise attrition

Business travel impact

Overtime impact

Distance from home analysis

Years at company analysis

Years since last promotion analysis

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to perform the analysis before building the dashboard.

8.Dashboard KPIs

The Power BI dashboard includes several Key Performance Indicators (KPIs) that provide an overview of employee attrition.

KPI 1 – Total Employees

Shows the total number of employees included in the dataset.

KPI 2 – Attrition Count

Displays the number of employees who left the organization.

KPI 3 – Attrition Rate

Represents the percentage of employees who left the organization.

KPI 4 – Retained Employees

Displays the total number of employees who remained in the organization.

KPI 5 – Number of Departments

Shows the total departments available in the organization.

KPI 6 – Number of Job Roles

Displays the number of unique job roles.

These KPIs provide management with a quick overview of workforce health.

9. Dashboard Visualizations

The interactive Power BI dashboard contains multiple charts to help HR managers understand employee attrition.

Department Wise Attrition

Shows which departments experience the highest employee turnover.

Job Role Wise Attrition

Identifies job roles with the highest attrition.

Age Group Wise Attrition

Compares employee exits across different age groups.

Gender Wise Attrition

Analyzes attrition among male and female employees.

Marital Status Wise Attrition

Compares attrition among Single, Married, and Divorced employees.

Business Travel vs Attrition

Shows whether employees who travel frequently leave the organization more often.

Overtime vs Attrition

Analyzes the relationship between overtime and employee turnover.

Distance from Home Analysis

Studies whether employees living farther from the workplace are more likely to leave.

Education Field Analysis

Compares employee attrition across different educational backgrounds.

Promotion Analysis

Analyzes how the number of years since the last promotion affects employee attrition.

Tenure Analysis

Shows employee attrition based on years spent in the organization.

10.Key Findings

The analysis generated several important business insights.

Employee attrition is concentrated in a limited number of departments.

Certain job roles experience significantly higher turnover.

Employees working overtime leave more frequently.

Frequent business travel increases the likelihood of attrition.

Early-career employees have higher attrition than experienced employees.

Employees waiting many years for promotion show increased turnover.

Younger employees leave more frequently than senior employees.

Work-life balance plays an important role in employee retention.

These findings can help HR teams design targeted retention strategies.

11.Recommendations

Based on the analysis, the following recommendations are proposed:

Improve Work-Life Balance

Reduce unnecessary overtime and encourage flexible work arrangements.

Career Development

Provide clear career growth opportunities and regular promotions.

Employee Engagement

Conduct regular employee satisfaction surveys.

Department-Level Monitoring

Closely monitor departments with consistently high attrition.

Managerial Support

Improve communication between employees and managers.

Employee Recognition

Reward high-performing employees to increase job satisfaction.

Training Programs

Offer continuous learning opportunities for career growth.

HR Intervention

Focus on employees identified as high-risk through dashboard insights.

12.Business Impact

The dashboard provides valuable support for HR decision-making.

Benefits include:

Faster HR reporting

Data-driven workforce planning

Better employee retention strategies

Improved management decisions

Reduced recruitment costs

Higher employee satisfaction

Better workforce productivity

13. Future Scope

This project can be extended further using Machine Learning techniques.

Possible future improvements include:

Employee Attrition Prediction

Machine Learning Classification Models

Employee Risk Score

Interactive Web Dashboard using Streamlit

Real-time HR Analytics

AI-based Retention Recommendations

14. Conclusion

Employee attrition is one of the most critical challenges faced by organizations. Through this project, employee data from Palo Alto Networks was analyzed using Python and Power BI to understand workforce attrition patterns.

The dashboard successfully identifies departments, job roles, age groups, and workload factors associated with higher employee turnover. The interactive dashboard allows HR professionals to explore attrition using multiple filters and KPIs.

The insights generated through this analysis can support better workforce planning, improve employee engagement, and reduce future attrition.

This project demonstrates how data analytics can help organizations transform raw employee data into meaningful business insights.

15. References

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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Google Colab, GitHub